

Welcome

Welcome to the Cyclistic bike-share analysis case study! In this case study, you work for a fictional company, Cyclistic, along with some key team members. In order to answer the business questions, follow the steps of the data analysis process: Ask, Prepare, Process, Analyze, Share, and Act. Along the way, the Case Study Roadmap tables — including guiding questions and key tasks — will help you stay on the right path.

Scenario

You are a junior data analyst working on the marketing analyst team at Cyclistic, a bike-share company in Chicago. The director of marketing believes the company's future success depends on maximizing the number of annual memberships. Therefore, your team wants to understand how casual riders and annual members use Cyclistic bikes differently. From these insights, your team will design a new marketing strategy to convert casual riders into annual members. But first, Cyclistic executives must approve your recommendations, so they must be backed up with compelling data insights and professional data visualizations.

Characters and Teams

- **Cyclistic:** A bike-share program that features more than 5,800 bicycles and 600 docking stations. Cyclistic sets itself apart by also offering reclining bikes, hand tricycles, and cargo bikes, making bike-share more inclusive to people with disabilities and riders who can't use a standard two-wheeled bike. The majority of riders opt for traditional bikes; about 8% of riders use the assistive options. Cyclistic users are more likely to ride for leisure, but about 30% use the bikes to commute to work each day.
- **Lily Moreno:** The director of marketing and your manager. Moreno is responsible for the development of campaigns and initiatives to promote the bike-share program. These may include email, social media, and other channels.
- **Cyclistic marketing analytics team:** A team of data analysts who are responsible for collecting, analyzing, and reporting data that helps guide Cyclistic marketing strategy. You joined this team six months ago and have been busy learning about Cyclistic's mission and business goals—as well as how you, as a junior data analyst, can help Cyclistic achieve them.

- **Cyclistic executive team:** The notoriously detail-oriented executive team will decide whether to approve the recommended marketing program.

About the Company

In 2016, Cyclistic launched a successful bike-share offering. Since then, the program has grown to a fleet of 5,824 bicycles that are geotracked and locked into a network of 692 stations across Chicago. The bikes can be unlocked from one station and returned to any other station in the system anytime.

Until now, Cyclistic's marketing strategy relied on building general awareness and appealing to broad consumer segments. One approach that helped make these things possible was the flexibility of its pricing plans: single-ride passes, full-day passes, and annual memberships. Customers who purchase single-ride or full-day passes are referred to as casual riders. Customers who purchase annual memberships are Cyclistic members.

Cyclistic's finance analysts have concluded that annual members are much more profitable than casual riders. Although the pricing flexibility helps Cyclistic attract more customers, Moreno believes that maximizing the number of annual members will be key to future growth. Rather than creating a marketing campaign that targets all-new customers, Moreno believes there is a solid opportunity to convert casual riders into members. She notes that casual riders are already aware of the Cyclistic program and have chosen Cyclistic for their mobility needs.

Moreno has set a clear goal: Design marketing strategies aimed at converting casual riders into annual members. In order to do that, however, the team needs to better understand how annual members and casual riders differ, why casual riders would buy a membership, and how digital media could affect their marketing tactics. Moreno and her team are interested in analyzing the Cyclistic historical bike trip data to identify trends.

ASK

Three questions will guide the future marketing program:

1. How do annual members and casual riders use Cyclistic bikes differently?
2. Why would casual riders buy Cyclistic annual memberships?
3. How can Cyclistic use digital media to influence casual riders to become members?

Moreno has assigned you the first question to answer: **How do annual members and casual riders use Cyclistic bikes differently?**

Stakeholders

- Lily Moreno: Director of Marketing
- Marketing Analytics Team
- Cyclistic Executive Team

Business Task

Analyze data to gain insights into how users use Cyclistic's bikes differently by membership type and to identify trends based on Cyclistic's marketing strategy.

PREPARE

The dataset I use is available from the link below:

[The previous 12 months of Cyclistic trip data](#)

Are there issues with bias or credibility in this data? Does your data ROCCC?

- **Reliable:** Data comes from a reputable source (Motive International Inc.).
- **Original:** This is primary data directly from Cyclistic's operations, not from a third party.
- **Comprehensive:** It covers full features for providing a complete picture of usage patterns.
- **Current:** The 4 quarterly datasets from 2019 may be enough to reflect relevant trends.
- **Cited:** The data is publicly available under a license from Motivate International Inc.
- **Bias Concern:** The data may underrepresent certain groups, such as cash versus digital payments or variations across geographic regions.

For this analysis, I used Python to explore the data.

Dataset Overview

All 4 quarterly datasets consists of **3,818,004 rows** and covers trips recorded from **January 1, 2019, to December 31, 2019**. It includes **12 columns** with details about each trip:

- **trip_id**: Unique identifier for each trip
- **start_time**: Date and time when the trip started
- **end_time**: Date and time when the trip ended
- **bikeid**: Unique identifier for the bike used
- **tripduration**: Total duration of the trip
- **from_station_id**: ID of the station where the trip started
- **from_station_name**: Name of the station where the trip started
- **to_station_id**: ID of the station where the trip ended
- **to_station_name**: Name of the station where the trip ended
- **usertype**: User type (Subscriber or Customer)
- **gender**: User's gender (if available)
- **birthyear**: User's birth year (if available)

Data Cleaning Process

1. **Rename Columns** – Make column names consistent across all datasets.
2. **Convert Data Types** – Remove commas and convert **tripduration** to numbers.
3. **Fill Missing Values**
 - **Gender**: Fill missing values based on the male-to-female ratio.
 - **Birth Year**: Fill missing values using the median based on user type and quarter.
4. **Add New Columns**
 - **Age**: Calculate by subtracting birth year from 2020.
 - **day_of_week**: Extract from **start_time** to show the day of the week
 - **Age Group**: Categorize users into age groups.

5. **Merge Data** – Combine all four quarterly datasets into one.

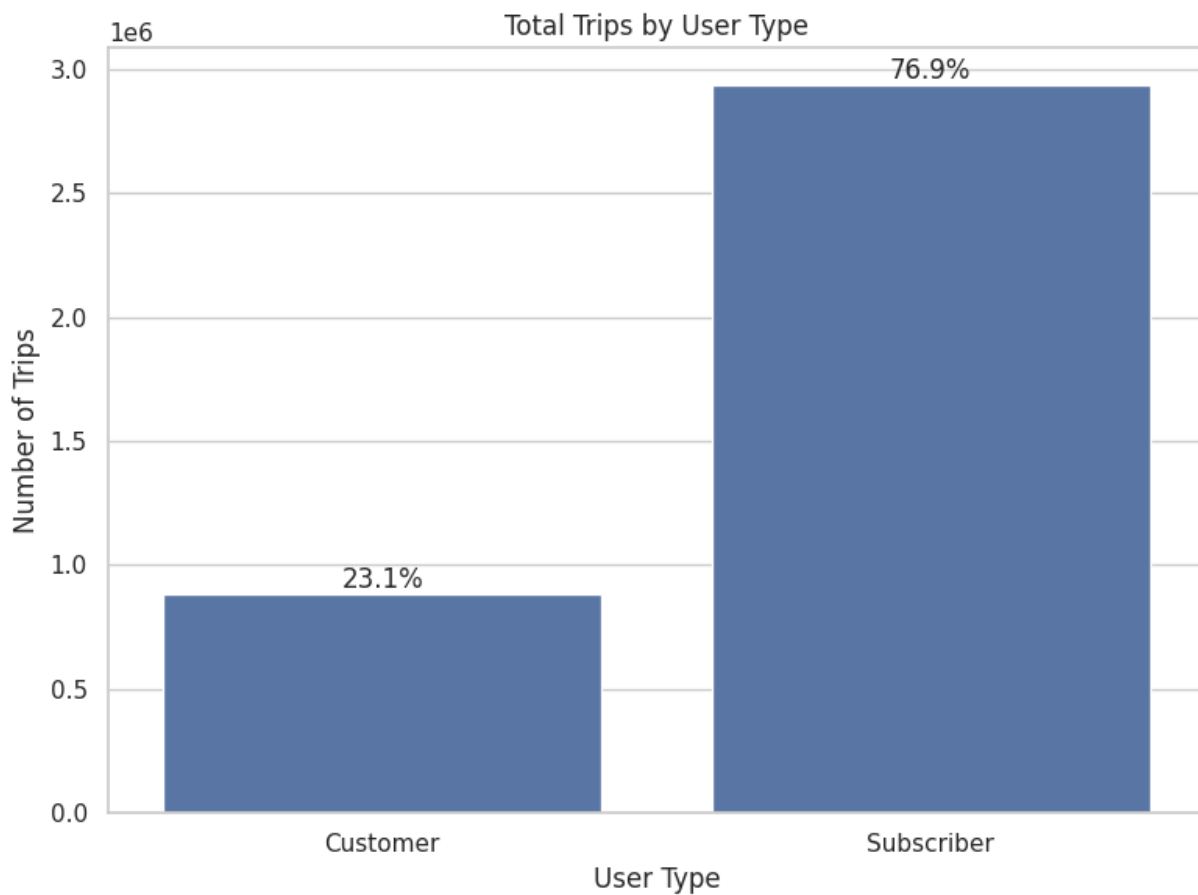
ANALYZE

In this section, we analyze how annual members (subscribers) and casual riders (customers) use the bike-share service differently. By examining Trip Frequency and Volume, Trip Duration, Station Usage patterns, Time of Use (Temporal Pattern), Demographic Distributions, and key differences, we aim to provide actionable insights for designing targeted marketing strategies.

Trip Frequency and Duration

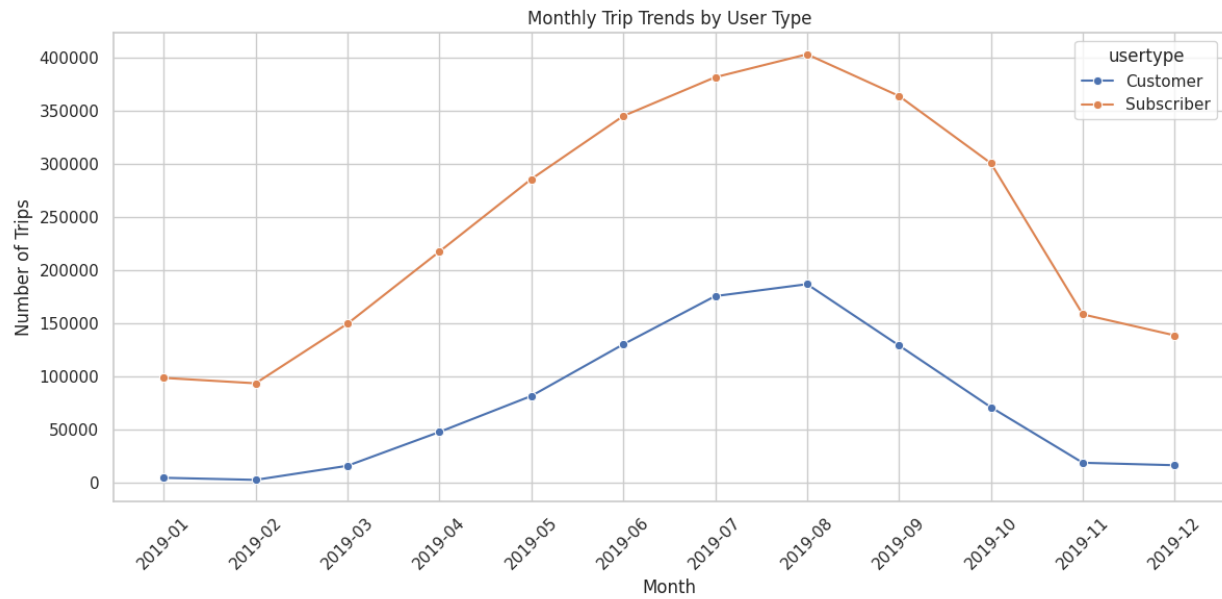
To better understand how annual members and casual riders differ, we first explored ride frequency and trip duration

Graph 1: Total Trips by User Type



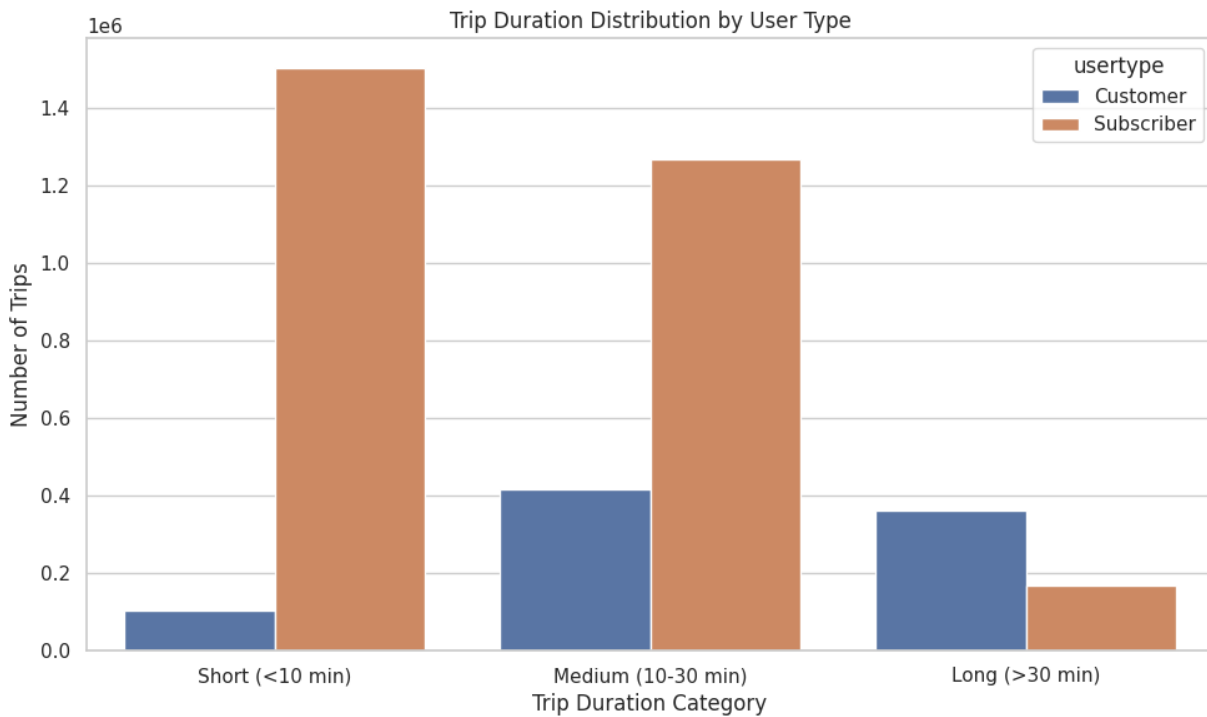
The number of trips for Subscriber are three time higher than Customer

Graph 2: Monthly Trip Trend by User Type



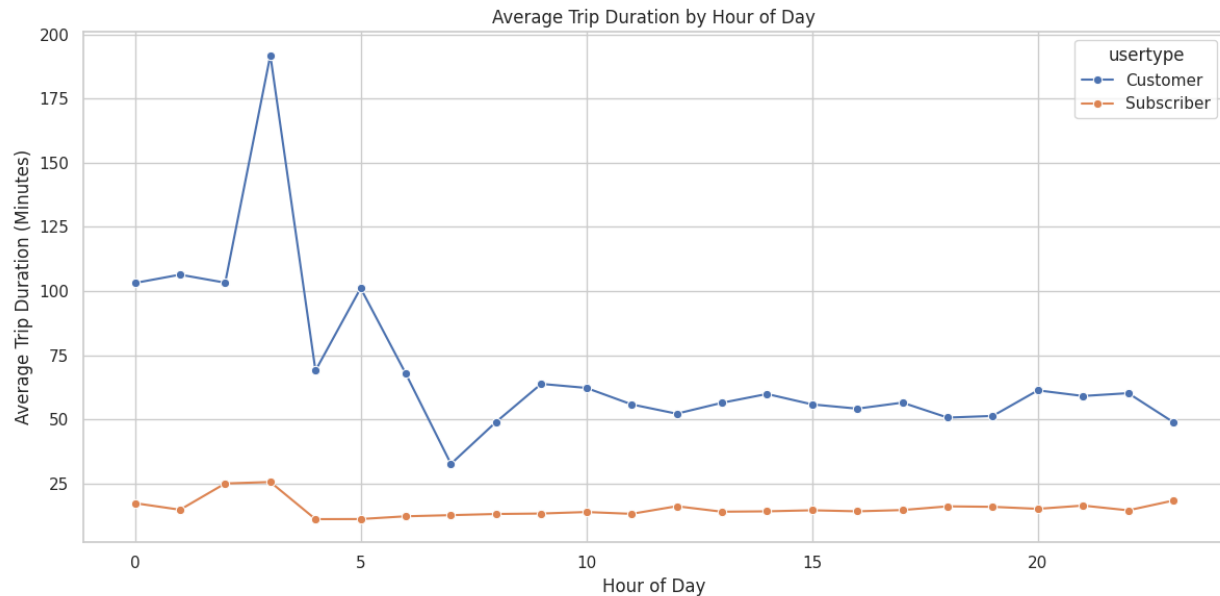
Both customers and subscribers follow similar seasonal trends, with ride counts peaking from May to September

Graph 3: Trip Duration Distribution by User Type



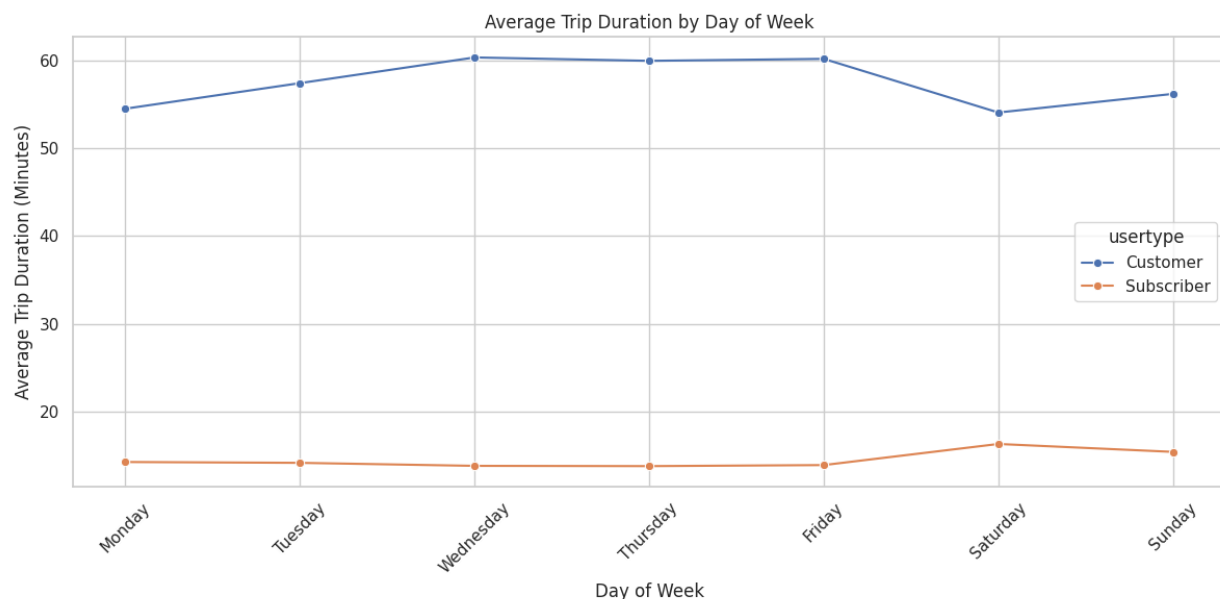
Most of the Subscriber's Trips are below 30 minutes while Customer' Trips took 10 to 30 minutes or even above 30 minutes

Graph 4: Average Trip Duration by hour of Day



Customers have higher average Trip Duration, especially in the morning around 3 am to 5 am.

Graph 5: Average Trip Duration by Day of Week



The same applies to Day of Week Where Customer also have higher average Trip Duration.

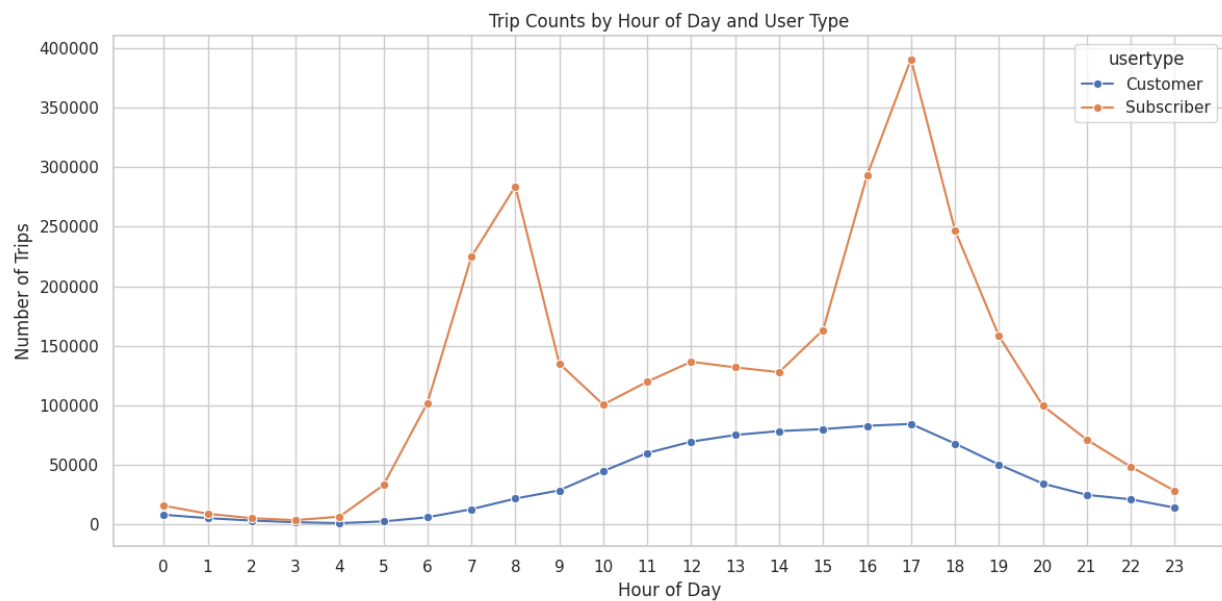
Combine all of the information above we have some Key Insights:

- Customers take longer trips on average, suggesting they may use bikes for leisure or tourism rather than routine travel.
- Both user types ride more frequently during warmer months, indicating a seasonal influence on bike usage.

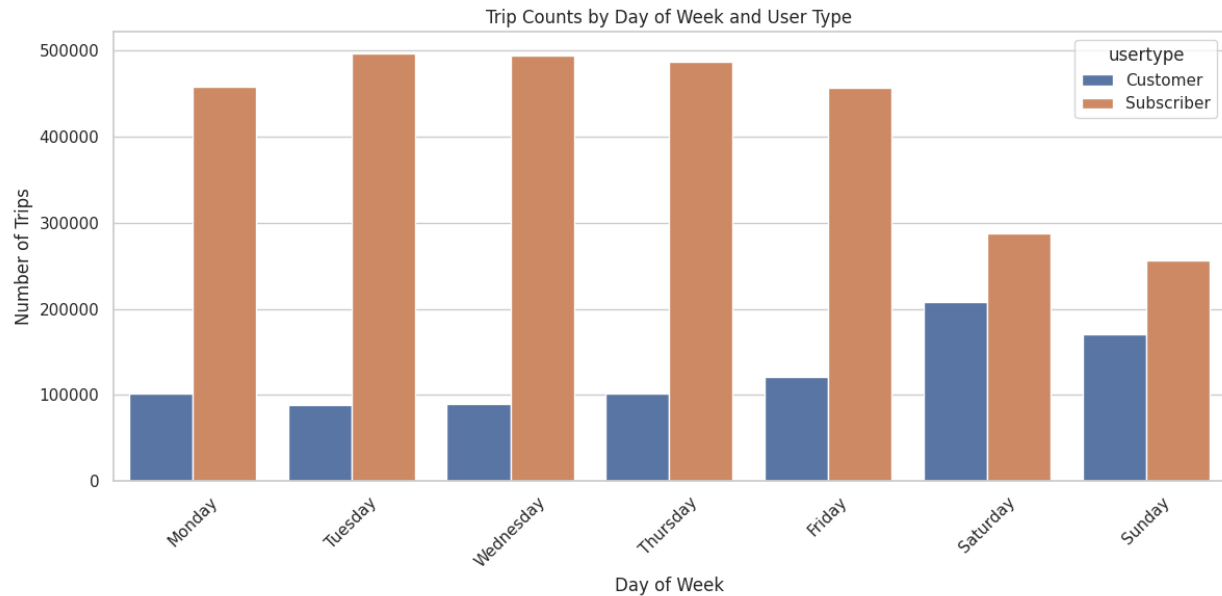
Time of Use (Temporal Patterns) Analysis

Next, we analyze Time of Use to identify when each user type uses bikes

Graph 6: Trip Count by hour of day



Graph 7: Trip Count by Day of Week



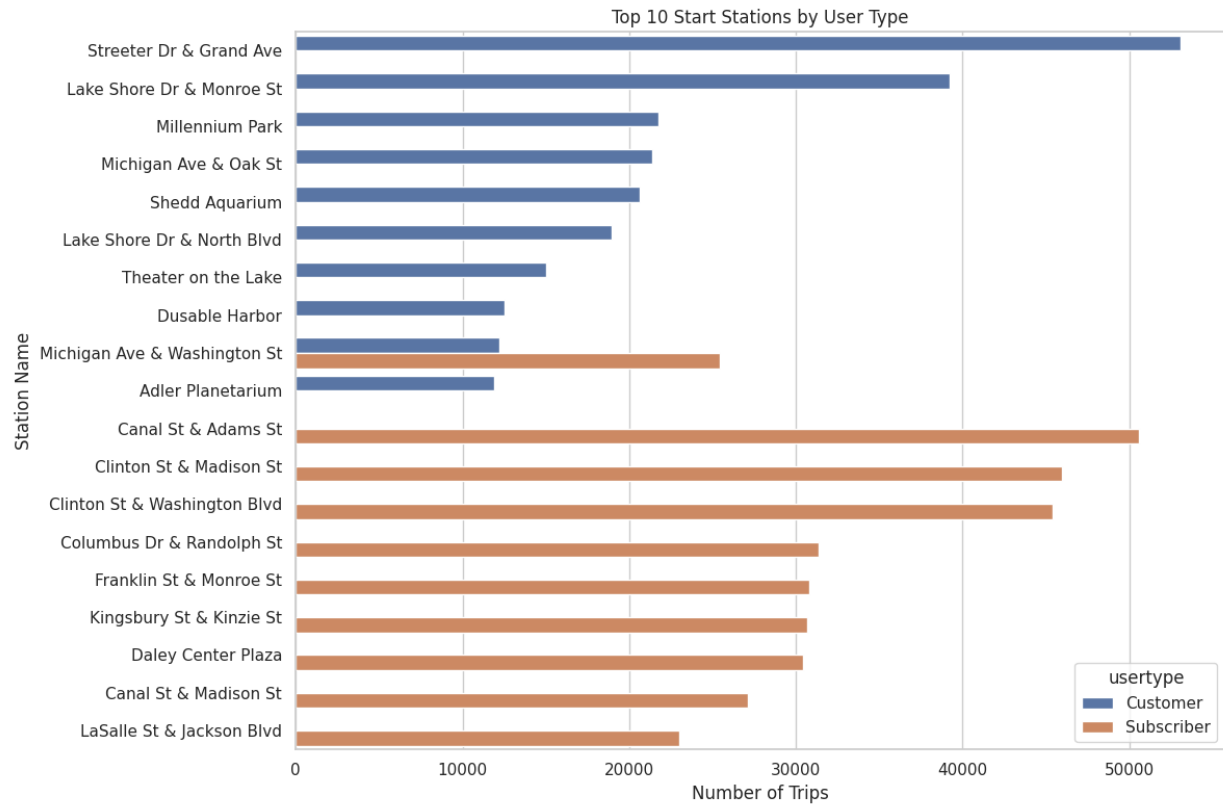
Key insight:

- Subscribers ride more consistently throughout the week peaking around morning (6 am to 9 am) and evening (3 pm to 7 pm)
- Customers, on the other hand, use bikes a lot more at the weekend, around afternoon and evening (from 12 pm to 7 pm)

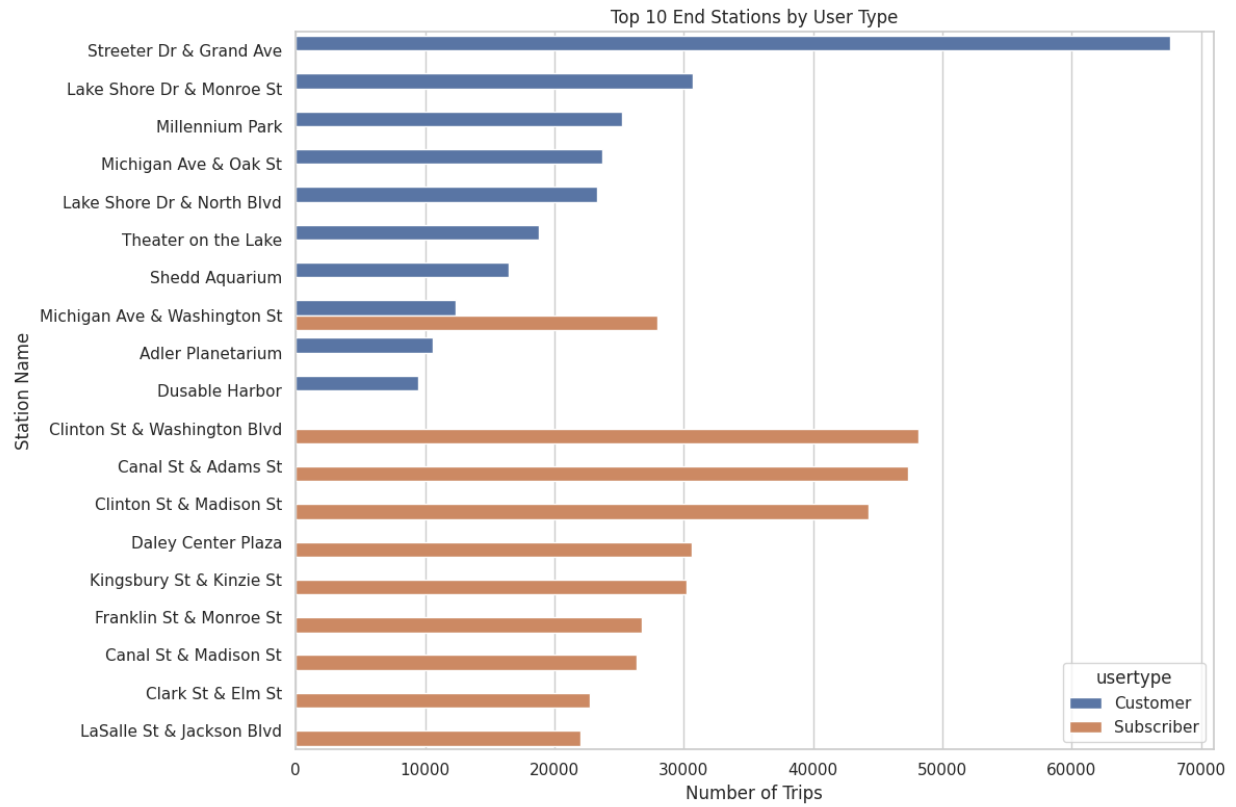
Station Usage (Spatial Patterns)

We analyzed station data to Understand where each user type starts and ends their trips.

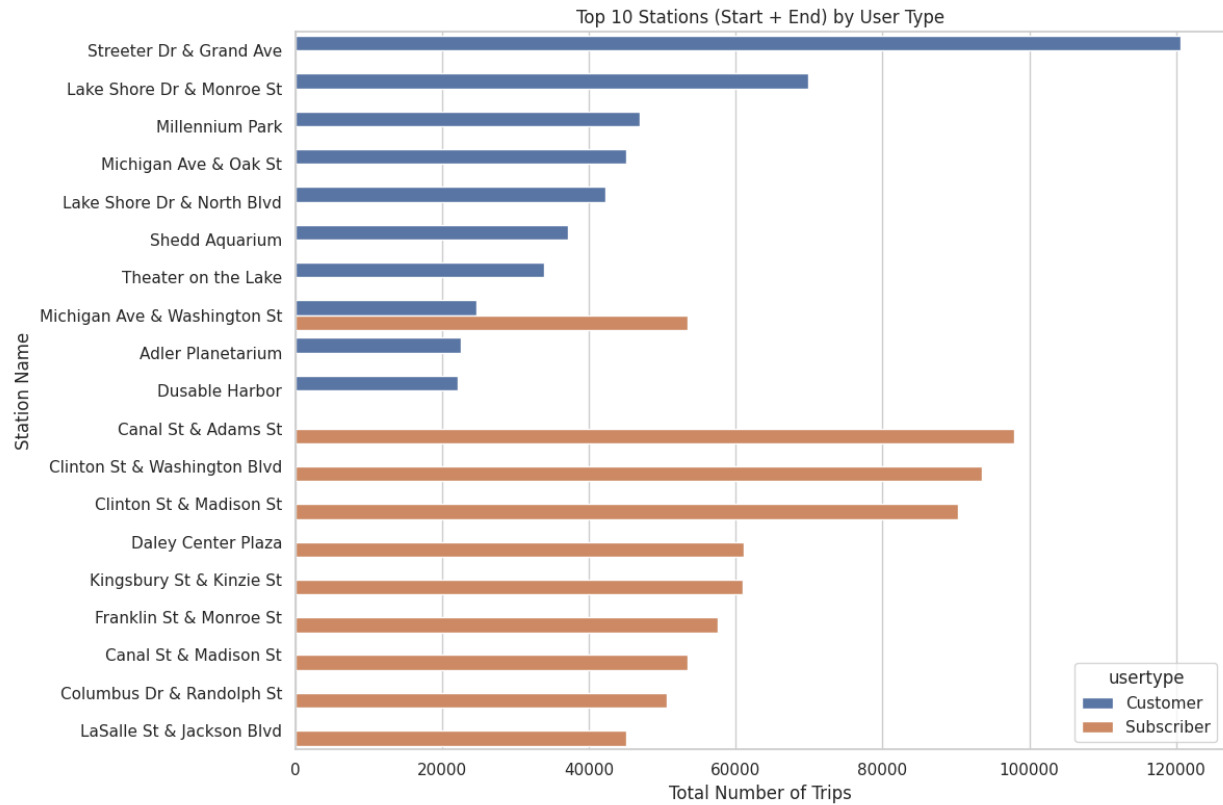
Graph 8: Top 10 Start Station



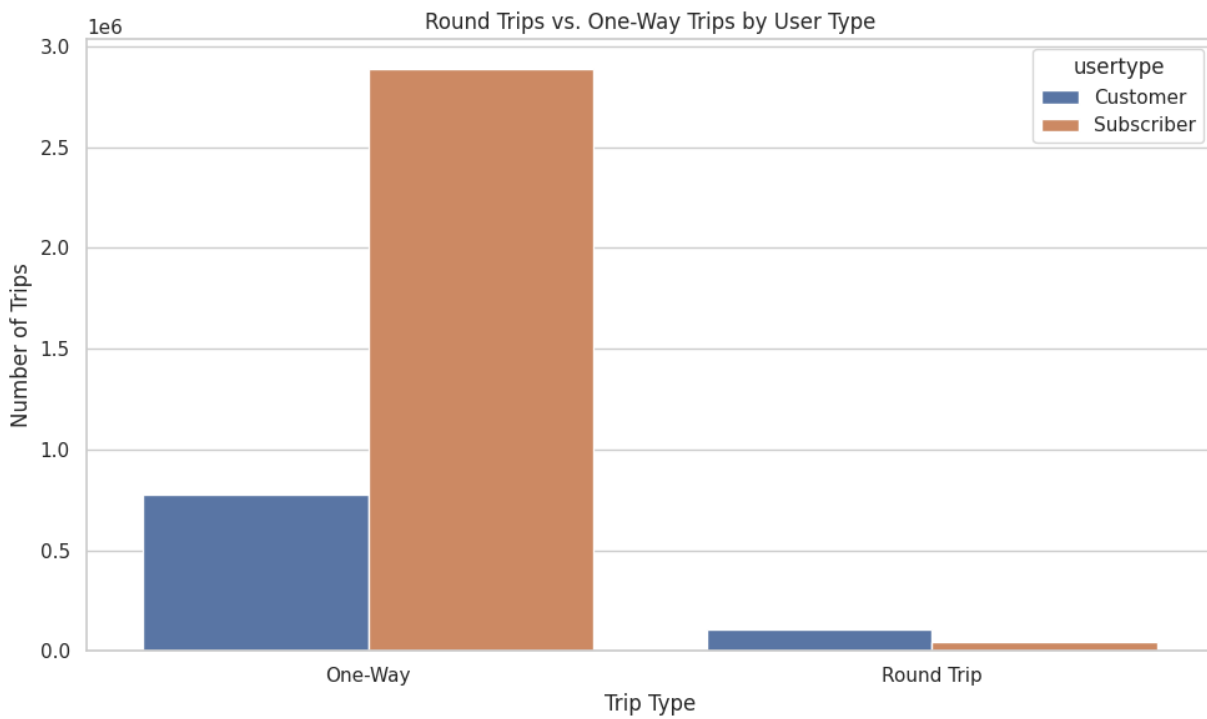
Graph 9: Top 10 End Station



Graph 10: Top 10 Start and End Station



Graph 11: Round Trips vs One-Way Trips



Key insight:

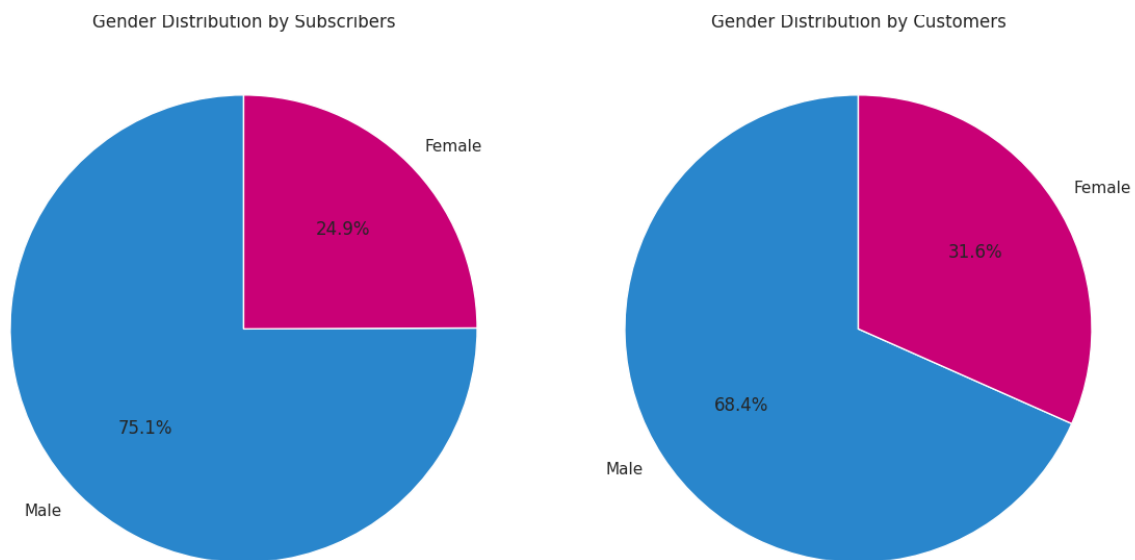
- Both Customer and Subscriber prefer to have One-Way trips with Subscriber is three time higher than Customer.
- Customers show a clear spike in rides at popular stations like Streeter Dr & Grand Ave, Millennium Park and Lake Shore Dr & Monroe St, which are tourist hotspots.
- Subscribers, on the other hand, show higher usage at stations likely tied to commuting routes (e.g., Canal St & Adam St, Clinton St, ...).

Demographic Analysis

We examined gender and age distributions to uncover demographic differences between customers and subscribers.

Gender Distribution

Graph 12: Pie Charts Showing Gender Distribution for Subscribers and Customers

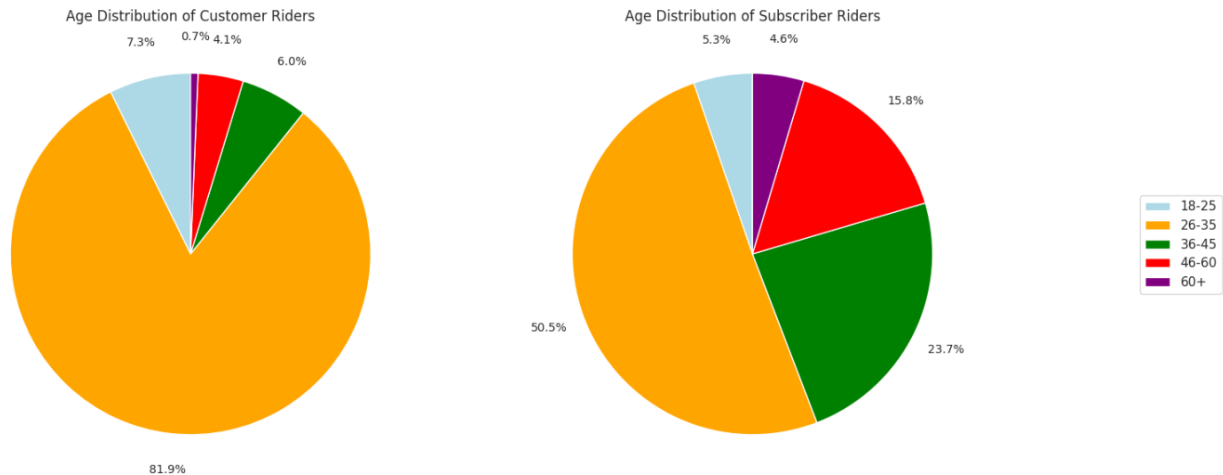


Key Insights:

- The proportion of female subscribers is 7% lower than that of female customers.
- This indicates that women are less likely to opt for annual memberships, possibly due to differing usage preferences or limited marketing appeal.

Age Distribution

Graph 13: Pie Chart Showing Age Distribution of Customers and Subscribers



Key Insights:

- The 26-35 age group dominates both user types, but it’s more pronounced among customers (81%) compared to subscribers (50.5%).
- Riders aged 36 and older are more likely to be subscribers than customers.

Younger riders (26-35) appear more inclined to use bikes casually, while older riders (36+) prefer subscriptions, possibly reflecting more established commuting habits.

Comparison of Customers vs. Subscribers

The table below summarizes the key differences between customers and subscribers based on the analysis:

Category	Customers	Subscribers
Trip Duration	Longer average trip durations	Shorter average trip durations
Ride Frequency	More rides on weekends	Consistent rides throughout the week
Seasonal Usage	Peaks in Summer and Fall	Peaks in Summer and Fall
Station Usage	High usage at tourist-heavy stations (e.g., Millennium Park)	High usage at commuting-related stations (e.g., Canal St & Adam St)
Gender Distribution	Higher proportion of females	Lower proportion of females
Age Distribution	Predominantly 26-35 years old (81%)	More evenly distributed, higher proportion of 36+ years old (50.5% in 26-35 group)

Key Insights:

- **Customers** are more likely to use Cyclistic's bikes for leisure or tourism, as evidenced by longer trips, weekend spikes, and activity at tourist-friendly stations.
- **Subscribers** demonstrate a pattern of regular, utilitarian use, with shorter, more frequent trips throughout the week and a broader age range, suggesting commuting or daily exercises.

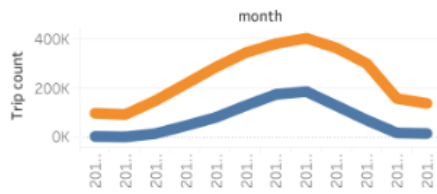
Conclusion

This analysis highlights distinct usage patterns between annual members and casual riders. Customers lean toward recreational use, influenced by weather and weekend availability, while subscribers exhibit consistent, practical usage tied to routines like commuting. These insights provide a foundation for crafting marketing strategies to convert casual riders into annual members by addressing their specific needs and preferences.

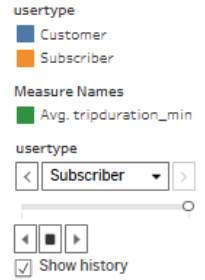
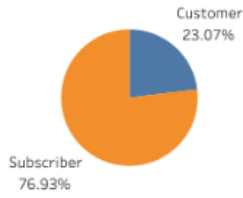
Share & Act

I used Tableau to recreate graphs, you can view it [here](#).

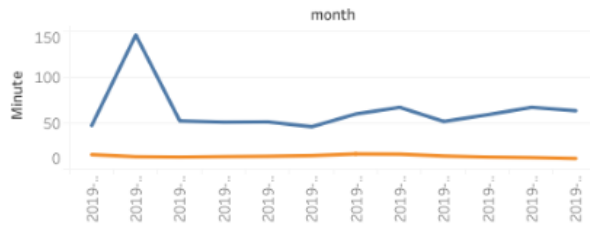
Total Trip



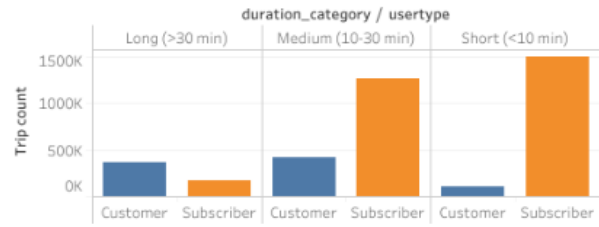
Trip percentage by usertype



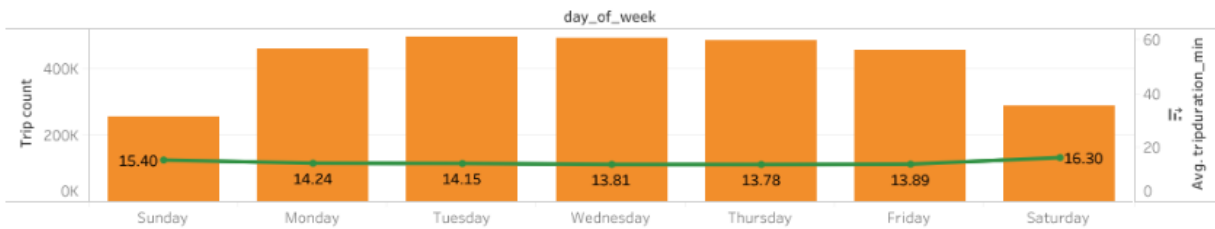
Avg Trip Duration (mins)



Duration Category



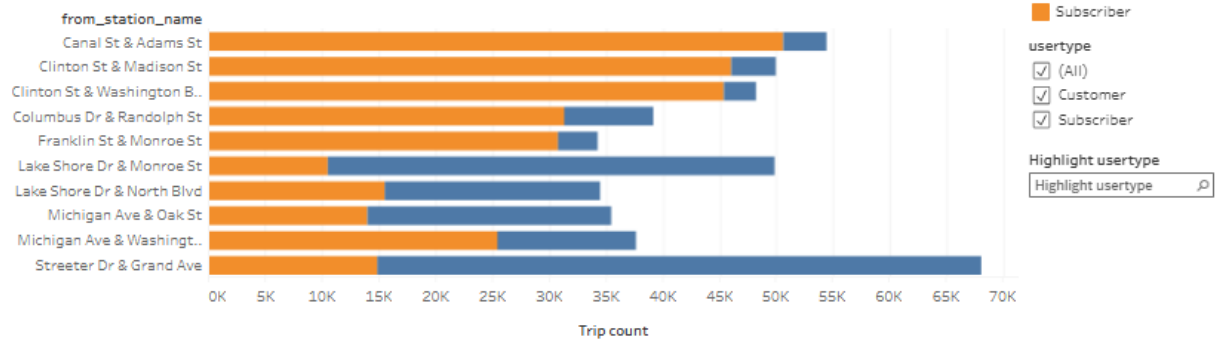
Weekly Ride Volume & Average Duration - Subscriber



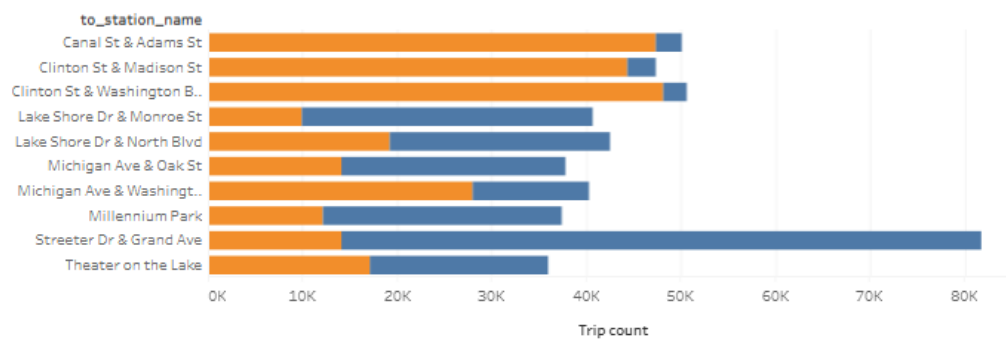
Station DashBoard

[Home](#)[Other Insight](#)

Station Usage (Start Station)

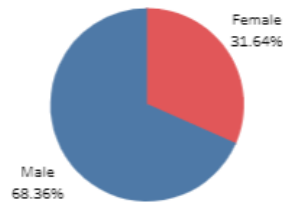


Station Usage (End Station)

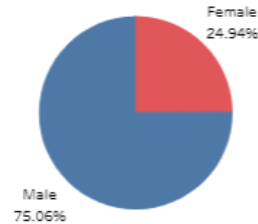


Other Insights

Customer's Gender



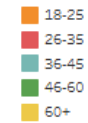
Subscriber's Gender



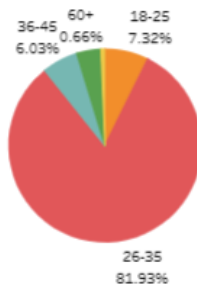
gender



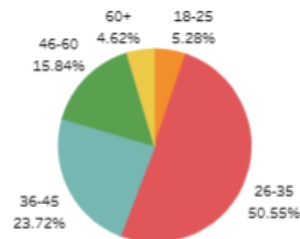
age_group



Customer's Age Group



Subscriber's Age Group



Recommendations

1. Annual Plan for Weekday Commuters: Annual subscription with unlimited weekday rides for commuters

- **Problem Solved:** Subscribers ride more on weekdays, indicating commuting patterns. This plan encourages casual riders who commute to switch from single-ride or day passes to annual memberships

2. Annual Plan for Summer & Autumn Only: Seasonal subscription for Q2 and Q3, priced lower than full-year plans.

- **Problem Solved:** Targets casual riders active in warmer months (Summer, Fall)

3. Plan Only for Weekends: Unlimited weekend rides with discounts for longer trips.

- **Problem Solved:** Casual riders predominantly use bikes on weekends at tourist-heavy stations like Millennium Park. This plan targets their recreational habits, making membership more attractive without a full-week commitment.

4. Discounts for Women on Special Days: Offer subscription discounts on days like International Women's Day, promoted through targeted digital campaigns.

- Problem Solved: Addresses 7% lower female subscriber proportion.